16/6/26 S - III

Total no. of Pages: 2

Roll no. ....

Degree: B.Tech. Semester: 3 Course work

MID-SEMESTER EXAMINATION, June 2026

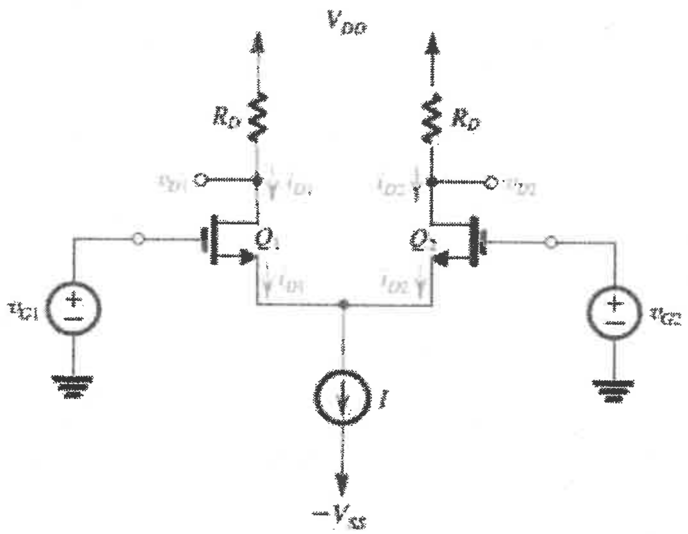
Course Title: Microelectronics

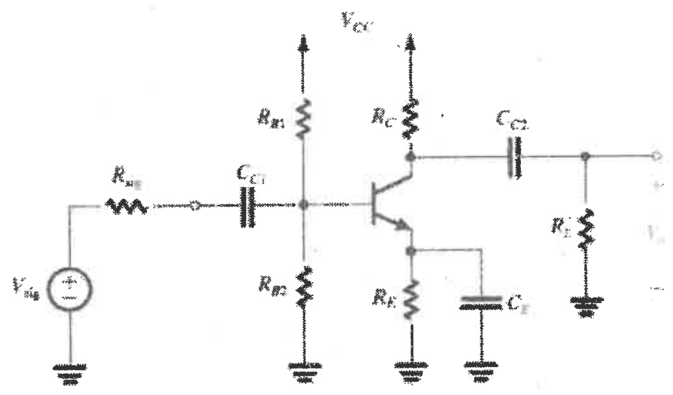
Course Code: ECECC07/VTECC304

Duration: 1:30 Hours

Max. Marks: 20/15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	For the BJT differential amplifier shown in Fig. 1(a), derive the expression for differential output voltage v_{od} in terms of the differential input voltage $v_{id=VG1-VG2}$.  Fig. 1	3/2	CO1
1b	Draw the small circuit equivalent circuit of the differential amplifier shown in Fig.1	1	CO1
2a	A MOS differential pair is biased with a tail current of 600 μA. Determine the drain currents when the differential input voltage is sufficiently large to completely steer the tail current.	3/2	CO1
2b	A two-stage amplifier has an overall gain of 240 V/V and first-stage gain of 15. Determine the gain of the second stage.	1	CO1

3a	A Darlington pair has $\beta_1 = 120$and $\beta_2 = 80$. Determine the overall current gain and input current required to drive a load current of 20 mA.	3/2	CO2
3b	Define Common-Mode Rejection Ratio (CMRR).	1	CO2
4a	A common-source amplifier has $g_m = 5\ mS$, $R_D = 12\ k\Omega$, $R_{sig} = 10\ k\Omega$, $C_{gd} = 2\ pF$, and $A_v = -30$. Determine the Miller input capacitance.	3/2	CO2
4b	Draw the high-frequency equivalent circuit of a cascode amplifier and explain how the Miller effect is minimized.	1	CO2
5a	For the common emitter amplifier shown in Fig. 2, derive an expression for the overall lower 3-dB cut-off frequency (f_L)using the Short-Circuit Time Constant (SCTC) method. Clearly state all assumptions made during the derivation.  Fig. 2	3/2	CO3
5b	Draw the small circuit equivalent circuit of the differential amplifier shown in Fig.2.	1	CO3